



1. useState – React Hook



SYNTAX

```
const [order, setOrder] = useState(initialValue);
```

- order → current value
- setOrder → function to update value
- initialValue → the default starting value



CODE EXAMPLE

```
// 📦 Import useState
import React, { useState } from 'react';

// ⚙️ Functional Component <Order>
function Order() {
  // Initial value: 0
  const [order, setOrder] = useState(0);

  return (
    <div>
      <h1>Order: {order}</h1>

      <button onClick={() => setOrder(order + 1)}>Add</button>
      <button onClick={() => setOrder(order - 1)}>Minus</button>
      <button onClick={() => setOrder(0)}>Reset</button>
    </div>
  );
}

export default Order;
```

✅ **Note:** `setOrder(Order + 1)` should be `setOrder(order + 1)` (JavaScript is case-sensitive)

Summary

- `useState` lets you manage **reactive state**
- Updating the state using `setOrder()` will **re-render** the component
- Perfect for counters, forms, toggles, and input tracking

2. `useEffect` – React Hook

SYNTAX

```
useEffect(() => {  
  // Side-effect logic here  
}, [dependencies]);
```

- Runs after component renders
- Re-runs when values in `[dependencies]` change
- If `[]` is empty → runs **only once** (on mount)

EXPLAIN

- `useEffect` is used to handle **side effects** like:
 - ✅ API calls
 - ✅ Timers
 - ✅ Event listeners
 - ✅ DOM updates
- Keeps your UI and logic **in sync** with outside systems



Real-Life Analogy: Hotel Order Example

Imagine you're in a hotel:

- A **customer** enters (component loads)
- A **waiter** (useEffect) notices
- He takes the **order** (state)
- Then runs to the **kitchen** (API call or timer) to prepare the food
- If **order changes**, he goes again
- When the customer leaves, the **table is cleaned** (cleanup function)



useEffect = Waiter who watches and reacts.



CODE EXAMPLE

```
// 📦 Import useEffect & useState
import React, { useState, useEffect } from 'react';

function UserData() {
  const [users, setUsers] = useState([]);

  useEffect(() => {
    // 🕒 Side effect: fetch user data from API
    fetch("https://jsonplaceholder.typicode.com/users")
      .then(res => res.json())
      .then(data => setUsers(data));
  }, []); // Empty array = run only once on mount

  return (
    <div>
      <h2>👤 User List</h2>
      {users.map(user => (
        <p key={user.id}>{user.name}</p>
      ))}
    </div>
  );
}

export default UserData;
```



CLEANUP EXAMPLE

```
useEffect(() => {
  const interval = setInterval(() => {
    console.log("🕒 Tracking...");
  }, 1000);

  return () => clearInterval(interval); // 🧹 Cleanup when component unmounts
}, []);
```

Summary

Concept	Meaning
Side Effect	Something that affects outside the UI
Empty []	Runs once when component mounts
[value]	Runs when value changes
Return function	Used for cleanup (like timers/events)

 `useEffect` = “Do this after render, and maybe again if something changes.”

3. useContext – React Hook

SYNTAX

```
const value = useContext(MyContext);
```

- Used to **access shared global data**
- Avoids **prop drilling** (no need to pass data through every level)
- Requires a **Context Provider**

EXPLAIN

- `useContext` lets multiple components **access the same data** without passing it manually
- Ideal for **themes, authentication, language settings, user data** etc.



Real-Life Analogy: The Wi-Fi Board Example

Imagine you're running a company.

- The **boss** writes the Wi-Fi password on a whiteboard
- Every **employee** just reads from the board
- No one asks the boss again and again!

💡 That's `useContext` !



HOW TO USE



1. Create the Context

```
import { createContext } from 'react';

export const WifiContext = createContext();
```



2. Wrap Components with Provider

```
import React, { useState } from 'react';
import { WifiContext } from './WifiContext';
import App from './App';

function WifiProvider() {
  const [wifiPassword] = useState("Coffee@123");

  return (
    <WifiContext.Provider value={wifiPassword}>
      <App />
    </WifiContext.Provider>
  );
}

export default WifiProvider;
```

✓ 3. Access in Any Child Component

```
import React, { useContext } from 'react';
import { WifiContext } from './WifiContext';

function Employee() {
  const wifi = useContext(WifiContext); // 🧠 Getting value from board

  return (
    <div>
      <h3>📶 Wi-Fi Password: {wifi}</h3>
    </div>
  );
}

export default Employee;
```

🧠 Summary

Concept	Meaning
createContext()	Creates shared global context
.Provider	Sends the value (like boss writing on board)
useContext()	Reads the value anywhere (like employee reading)
Use Case	Theme, auth, language, global user data

🧠 useContext = Global whiteboard.

One person writes, everyone can read. No need to pass papers down the line!



4. useRef – React Hook



SYNTAX

```
const ref = useRef(initialValue);
```

- Stores a **mutable reference** (like a box)
- Does **not cause re-render** when updated
- Commonly used for:
 -  Accessing DOM nodes
 -  Storing timers
 -  Holding previous values



EXPLAIN

- `useRef` is like a **secret pocket** inside your component.
- You can store **anything**, and React won't re-render when it changes.
- It's often used to **directly control a DOM element** (like focusing input).



Real-Life Analogy: The Calculator-in-Hand Example

Imagine you're a **cashier**.

- You keep a **calculator in your hand**
- You don't tell the manager every time you use it (no re-render)
- You use it when needed (DOM access), but it's **not part of your official report**



That's `useRef` .



CODE EXAMPLE – Focusing an Input

```
import React, { useRef } from 'react';

function FocusInput() {
  const inputRef = useRef(null); // 🟡 Like holding calculator

  const handleFocus = () => {
    inputRef.current.focus(); // 🔑 Direct DOM access
  };

  return (
    <div>
      <input ref={inputRef} type="text" placeholder="Enter your name" />
      <button onClick={handleFocus}>🔑 Focus Input</button>
    </div>
  );
}

export default FocusInput;
```



CODE EXAMPLE – Store a Timer ID

```
const timerRef = useRef();

useEffect(() => {
  timerRef.current = setInterval(() => {
    console.log("🕒 Tick");
  }, 1000);

  return () => clearInterval(timerRef.current); // Clean up
}, []);
```

Summary

Concept	Meaning
<code>useRef()</code>	Stores a mutable reference
<code>.current</code>	The actual stored value
DOM access	<code>ref.current.focus()</code>
No re-render	Changing <code>.current</code> doesn't trigger re-render
Use case	DOM, timers, previous values, imperatives

 `useRef` = "Calculator in your hand — helps you act fast, but doesn't change the report (UI)."

5. useMemo – React Hook

SYNTAX

```
const memoizedValue = useMemo(() => computeExpensiveValue(dep), [dep]);
```

- Runs the function **only when dependencies change**
-  Caches (remembers) the result
-  Prevents **unnecessary recalculations**

EXPLAIN

- `useMemo` is used to **optimize performance**
- Perfect when you have **heavy calculations**
- It **skips re-execution** if input hasn't changed



Real-Life Analogy: Restaurant Bill Calculator

Imagine you're the **cashier** at a hotel.

- A group orders 20+ items
- You calculate the full bill (takes time)
- If they **add one more** item, you recalculate
- But if **nothing changed**, you just reuse the old total!

💡 That's `useMemo` — **calculate only when needed**.



CODE EXAMPLE – Expensive Calculation

```
import React, { useMemo, useState } from 'react';

function BillCalculator() {
  const [items, setItems] = useState([100, 200, 150]);
  const [tax, setTax] = useState(0.18);

  const total = useMemo(() => {
    console.log("💰 Calculating total...");
    return items.reduce((sum, price) => sum + price, 0) * (1 + tax);
  }, [items, tax]); // 🔄 Only recalculate if items or tax change

  return (
    <div>
      <h3> 📄 Total Bill: ₹{total.toFixed(2)}</h3>
    </div>
  );
}
```

```
export default BillCalculator;
```

Summary

Concept	Meaning
useMemo()	Caches a value based on dependencies
Runs only when...	Dependencies change
Best use case	Expensive calculations, filtering, sorting etc.
Return value	Reused on future renders unless changed

 `useMemo` = Like saving your restaurant bill total and **only rechecking if someone changes the order**.

6. useCallback – React Hook

SYNTAX

```
const memoizedFn = useCallback(() => {  
  // callback logic here  
}, [dependencies]);
```

- Returns a **memoized version of a callback function**
- Only **recreates the function** if dependencies change
- Used to prevent **unnecessary re-renders** or **function re-creation**

EXPLAIN

- In React, functions are **recreated on every render** by default

- `useCallback` helps to **avoid passing a new function each time**
- Useful when you pass functions to **child components**, and want to avoid unnecessary re-renders

Real-Life Analogy: Coffee Button in a Hotel

Imagine you're in a hotel room with a **coffee machine**.

- You press the  **"Make Coffee"** button (function)
- Hotel staff don't need to reset it every second — unless you change **coffee powder** or **water**
- If nothing changed, the same button works again!

💡 That's `useCallback` — it **reuses the function**, unless something it depends on changes.



CODE EXAMPLE – Button Callback with Memoization

```
import React, { useState, useCallback } from 'react';

function CoffeeOrder() {
  const [count, setCount] = useState(0);
  const [orderNote, setOrderNote] = useState("");

  const handleOrder = useCallback(() => {
    console.log("☕ Order placed with note:", orderNote);
  }, [orderNote]); // Only recreates when orderNote changes

  return (
    <div>
      <input
        type="text"
        placeholder="Add order note"
        onChange={(e) => setOrderNote(e.target.value)}
      />
      <button onClick={handleOrder}>Place Order</button>
      <p>Orders placed: {count}</p>
      <button onClick={() => setCount(count + 1)}>+1</button>
    </div>
  );
}

export default CoffeeOrder;
```



When to Use?

- Passing **callbacks to optimized child components** (`React.memo`)
- Inside custom hooks
- When working with `useEffect` or event listeners

Summary

Concept	Meaning
useCallback()	Memoizes a function
Avoids re-creation	Only changes if dependencies change
Use case	Pass function to child without triggering re-render
Compare with	<code>useMemo</code> for values, <code>useCallback</code> for functions

 `useCallback` = Reuse your coffee button.

Don't remake it unless the ingredients (dependencies) change 

7. `useReducer` – React Hook (State Manager)

SYNTAX

```
const [state, dispatch] = useReducer(reducerFunction, initialState);
```

- `state` → current state value
- `dispatch(action)` → used to update state
- `reducerFunction(state, action)` → function that returns the new state
- `initialState` → starting state

EXPLAIN

- `useReducer` is like **a mini Redux** inside a component
- Useful when:
 - ☒ State logic is **complex**
 - ☒ You have **multiple sub-states**
 - ☒ Updates depend on **previous state**
- Cleaner alternative to `useState` for complex logic

Real-Life Analogy: Hotel Order Management

Imagine you run a hotel kitchen:

- You have a **notebook** (state) for order counts
- The waiter gives you **instructions** like "Add 1", "Reset", "Remove 1"
- Based on the instruction (`action.type`), you **update the notebook**

💡 That's `useReducer` — a central logic to handle many actions.



CODE EXAMPLE – Order Counter

```
import React, { useReducer } from 'react';

// Reducer function - handles all actions
function orderReducer(state, action) {
  switch (action.type) {
    case 'increment':
      return { count: state.count + 1 };
    case 'decrement':
      return { count: state.count - 1 };
    case 'reset':
      return { count: 0 };
    default:
      return state;
  }
}

const initialState = { count: 0 };

function HotelOrder() {
  const [state, dispatch] = useReducer(orderReducer, initialState);

  return (
    <div>
      <h2>📦 Orders: {state.count}</h2>

      <button onClick={() => dispatch({ type: 'increment' })}>Add Order</button>
      <button onClick={() => dispatch({ type: 'decrement' })}>Cancel Order</button>
      <button onClick={() => dispatch({ type: 'reset' })}>Reset</button>
    </div>
  );
}

export default HotelOrder;
```

Summary

Concept	Meaning
<code>useReducer()</code>	Hook for managing complex state
<code>state</code>	Current value of the state
<code>dispatch()</code>	Function to send actions
<code>reducer()</code>	Pure function to update state based on action
Use case	Forms, toggles, carts, auth, multi-step logic

 `useReducer` = Hotel notebook for managing different order instructions with one central rulebook (the reducer)!

8. `useImperativeHandle` – React Hook (Custom Ref Control)

SYNTAX

```
useImperativeHandle(ref, () => ({
  method1,
  method2,
}), [dependencies]);
```

- Used with `forwardRef`
- Lets you **expose specific functions or values to parent**
- Parent can **call methods inside the child** like `.focus()` , `.reset()` , etc.

EXPLAIN

Normally, `ref` is used to access DOM nodes.

But sometimes you want to **expose custom functions from a child component to its parent**.

That's what `useImperativeHandle` does:

- Gives the parent **controlled access** to child functions
- Used rarely, but powerful in cases like:
 - Input focus
 - Form reset
 - Custom animations
 - Modal open/close

Real-Life Analogy: Hotel Room Service Bell

Imagine you're staying in a hotel.

- You don't go to the kitchen (child) every time.
- You just press the **service bell** from your room (ref).
- Only certain actions are allowed: "Bring Water", "Make Coffee".

 That's `useImperativeHandle` — the **service bell** gives parent access to a few allowed actions inside the child component.



CODE EXAMPLE – Focus Input from Parent



Child: InputBox.jsx

```
import React, { useRef, forwardRef, useImperativeHandle } from 'react';

const InputBox = forwardRef((props, ref) => {
  const inputRef = useRef();

  useImperativeHandle(ref, () => ({
    focusInput() {
      inputRef.current.focus(); // Exposed method
    },
    clearInput() {
      inputRef.current.value = '';
    }
  }));

  return <input ref={inputRef} placeholder="Type something..." />;
});

export default InputBox;
```



Parent: App.jsx

```
import React, { useRef } from 'react';
import InputBox from './InputBox';

function App() {
  const inputBoxRef = useRef();

  return (
    <div>
      <InputBox ref={inputBoxRef} />
      <button onClick={() => inputBoxRef.current.focusInput()}>Focus Input</button>
      <button onClick={() => inputBoxRef.current.clearInput()}>Clear Input</button>
    </div>
  );
}

export default App;
```



Summary

Concept	Meaning
useImperativeHandle	Customize what a parent can access via ref
forwardRef()	Required to forward the ref to child
.focus() , .clear()	Examples of methods exposed to parent
Use cases	Form control, modals, animations, focus handling



useImperativeHandle = Like pressing a hotel bell that only triggers **approved services** from the child (input, modal, etc.)

9. useEffect – React Hook (Sync DOM Effect)

SYNTAX

```
useEffect(() => {  
  // sync DOM effect here  
}, [dependencies]);
```

- Runs **after the DOM is updated**, but **before the browser paints**
- Runs synchronously → **blocks painting**
- Use only when you **must read layout or measure DOM**

EXPLAIN

- Very similar to `useEffect`, but **executes earlier**
- Use when:
 - You need to **measure DOM elements**
 - Adjust styles before screen shows
 - Fix UI shifts (prevent flicker)

Real-Life Analogy: Adjusting Table Setup Before Guests See It

Imagine you're in a restaurant:

- A waiter quickly aligns the tablecloth straight **before** the customer walks in.
- If he adjusts it **after** the customer sees it → awkward!

💡 That's `useLayoutEffect` .

Used when you want to **do something before the screen shows it** to the user.



CODE EXAMPLE – Auto Scroll Before Paint

```
import React, { useEffect, useRef } from 'react';

function ScrollBox() {
  const boxRef = useRef();

  useEffect(() => {
    // Scroll to bottom before screen is painted
    boxRef.current.scrollTop = boxRef.current.scrollHeight;
  }, []);

  return (
    <div
      ref={boxRef}
      style={{
        height: "150px",
        width: "300px",
        overflow: "auto",
        border: "1px solid gray",
      }}
    >
      <p>Item 1</p>
      <p>Item 2</p>
      <p>Item 3</p>
      <p>Item 4</p>
      <p>Item 5</p>
      <p>Item 6</p>
      <p>Item 7</p>
      <p>Item 8</p>
      <p>Item 9</p>
      <p>Item 10</p>
    </div>
  );
}

export default ScrollBox;
```


Warning

- **Blocks paint** → avoid heavy logic inside
- Prefer `useEffect` unless you see layout glitches or need measurements
- Use only when **timing is critical**

Summary

Concept	Meaning
<code>useLayoutEffect</code>	Runs before paint, after DOM update
Synchronous?	 Yes
Use when?	Measuring layout, fixing flicker, sync position
Caution	Don't use for heavy tasks – can delay rendering

 `useLayoutEffect` = “Fix the table setup quickly before guests (users) walk in and see it!”

10. `useId` – React Hook (Unique ID Generator)

SYNTAX

```
const id = useId();
```

- Generates a **unique and stable ID string** for this component instance
- Same across client/server — important for SSR (Server-Side Rendering)

- Useful for inputs, labels, modals, etc.

EXPLAIN

- Every time a component is created, `useId()` gives it a **unique ID**
- You can **append** strings to make it readable or assign it to form elements
- Prevents **ID collisions** in the DOM (duplicate IDs = **✗** bad for accessibility)

Real-Life Analogy: Room Numbers in a Hotel

Imagine a hotel assigns room numbers.

- Every guest gets a **unique room number**
- Two guests can't have **room 101** — that causes confusion!
- So React gives each guest (component) their **own stable ID**

💡 That's `useId` — unique room numbers for each element that need to be identified.

CODE EXAMPLE – Input & Label

```
import React, { useId } from 'react';

function NameInput() {
  const id = useId(); //  Unique ID

  return (
    <div>
      <label htmlFor={id}>Name:</label>
      <input id={id} type="text" placeholder="Enter your name" />
    </div>
  );
}

export default NameInput;
```



CODE EXAMPLE – Multiple IDs in a List

```
import React, { useId } from 'react';

function MenuList() {
  const id = useId();

  const items = ["Coffee", "Tea", "Juice"];
  return (
    <ul>
      {items.map((item, index) => (
        <li key={index} id={` ${id}-${index}`}>
          {item}
        </li>
      ))}
    </ul>
  );
}
```



Summary

Concept	Meaning
useId()	Generates unique, stable IDs for accessibility
SSR safe?	✅ Yes – same on server and client
Use case	Forms, labels, modals, lists
Format	Looks like: :r1: , :r2: , :r1-0

💡 useId = Give each guest (component) their own **room number**, so no one gets mixed up in the hotel 🏨

11. use – React 19 Hook (Handle Async Promises Directly)

SYNTAX

```
const data = use(promise);
```

- Accepts a **Promise**
- Suspends the component until the promise resolves
- Can be used inside **Server Components** and **Client Components** (with Suspense)
- **No need for** `useEffect` , `useState` , or even `async/await` in many cases!

EXPLAIN

- `use()` lets you **await promises directly** in React component body
- React **pauses rendering** until the promise resolves
- Helps **fetch data cleanly** without boilerplate

Real-Life Analogy: Order, Wait & Serve

Imagine you're at a restaurant.

- You order a biryani (promise)
- Waiter tells you: "Please wait until it's ready" (suspense)
- Once biryani is ready → waiter serves it (resolved promise)

 That's `use()` → it **waits silently**, and once data is ready, React continues rendering.



CODE EXAMPLE – Fetch Todo

✓ Works in components wrapped with `<Suspense>`

```
import React, { Suspense, use } from 'react';

const todoPromise = fetch("https://jsonplaceholder.typicode.com/todos/1")
  .then(res => res.json());

function Todo() {
  const todo = use(todoPromise); // ⌚ Suspends until resolved

  return (
    <div>
      <h2>✓ {todo.title}</h2>
    </div>
  );
}

export default function App() {
  return (
    <Suspense fallback={<p>Loading todo...</p>}>
      <Todo />
    </Suspense>
  );
}
```



CODE EXAMPLE – With Async Function

```
function fetchUser() {  
  return fetch("https://jsonplaceholder.typicode.com/users/1")  
    .then(res => res.json());  
}  
  
const userPromise = fetchUser();  
  
function Profile() {  
  const user = use(userPromise);  
  
  return <h1>Hello, {user.name}</h1>;  
}
```



Summary

Concept	Meaning
use()	Handles promises directly inside component
Suspense?	✅ Required to wrap component
Replaces	useEffect + useState combo for async data
Works in	React 19, with async-supported environments

💡 use() = “Place the order, wait silently behind the scenes, then React continues once it's served.”



React 19 Form & UX Hooks

12. useOptimistic – Instant UI Before Server Confirms

SYNTAX

```
const [optimisticValue, addOptimistic] = useOptimistic(value, updaterFn);
```

- value : actual source of truth
- addOptimistic(input, fn) : apply optimistic update (temporary)
- UI **instantly shows result** before server confirms

Real-Life Analogy: Instant Bill on POS Machine

Imagine ordering food:

- You pay → receipt prints instantly (optimistic)
- Actual bill from kitchen comes 5 secs later (real confirmation)

 useOptimistic = print receipt **before** confirmation

CODE EXAMPLE

```
'use client';
import { useState, useOptimistic } from 'react';

function Comments() {
  const [comments, setComments] = useState([]);
  const [optimisticComments, addOptimisticComment] = useOptimistic(
    comments,
    (current, newComment) => [...current, newComment]
  );

  const handleSubmit = async (formData) => {
    const comment = formData.get("comment");
    addOptimisticComment({ text: comment });

    await new Promise((res) => setTimeout(res, 1000)); // simulate server
    setComments(prev => [...prev, { text: comment }]);
  };

  return (
    <form action={handleSubmit}>
      <input name="comment" />
      <button type="submit">Post</button>
      <ul>
        {optimisticComments.map((c, i) => <li key={i}>{c.text}</li>)}
      </ul>
    </form>
  );
}
```

13. useFormStatus – Get Form Submit Status (Inside Form)

SYNTAX

```
const { pending, action } = useFormStatus();
```

- Tells if the **form is currently submitting**
- Used inside a `form` to **disable buttons**, show loading, etc.

Real-Life Analogy: Waiter Sending Order to Kitchen

- While waiter is sending your order (form submitting), you can't place another order
- Once done → order is enabled again

 `useFormStatus` = Track whether the form is **in progress**

CODE EXAMPLE

```
'use client';
import { useFormStatus } from 'react-dom';

function SubmitButton() {
  const { pending } = useFormStatus();

  return (
    <button type="submit" disabled={pending}>
      {pending ? "Sending..." : "Submit"}
    </button>
  );
}

function ContactForm() {
  async function handleSubmit(formData) {
    await new Promise(res => setTimeout(res, 2000)); // simulate delay
  }

  return (
    <form action={handleSubmit}>
      <input name="email" placeholder="Your email" />
      <SubmitButton />
    </form>
  );
}
```

14. useFormState – Manage Server Form State in Client

SYNTAX

```
const [state, formAction] = useFormState(actionFn, initialState);
```

- Helps manage **form state returned from server**

- Works with **Server Actions** (like validation result or status)

💡 Real-Life Analogy: OTP Verification

- You submit OTP form
- Server replies: ✅ Valid or ❌ Wrong
- Form remembers the result and shows message

💡 useFormState = Manage **form responses from server** easily

📦 CODE EXAMPLE

```
'use client';
import { useFormState } from 'react-dom';

async function verifyOtp(prevState, formData) {
  const otp = formData.get("otp");
  if (otp === "1234") return "✅ Verified";
  return "❌ Incorrect OTP";
}

function OtpForm() {
  const [message, formAction] = useFormState(verifyOtp, null);

  return (
    <form action={formAction}>
      <input name="otp" />
      <button type="submit">Verify</button>
      {message && <p>{message}</p>}
    </form>
  );
}
```

Summary Table

Hook	Purpose
useOptimistic	Show UI changes instantly , update later
useFormStatus	Know if form is submitting (loading)
useFormState	Manage server-side state of a form

💡 React 19 makes forms feel **native, fast, and smooth**, giving full control to both **UI and server interaction**.